

Adaptive Client Selection for Fair and Efficient Federated Learning in Non-IID Distributed Environments

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Abstract—Federated Learning (FL) facilitates cooperative training of models between distributed devices without violating data privacy. In practice, though, Non-Independent and Identically Distributed (Non-IID) data between clients is an outcome of unbiased global models, slow convergence, and unfair performance amongst the participants. The paper suggests a new adaptive client selection model which dynamically ranks clients with a lightweight composite score, relying on three local statistics, including dataset size: n_i , entropy of label distribution: H_i , and gradient divergence: D_i . The proposed procedure addresses the negative effects of statistical heterogeneity to a great extent, thanks to the emphasis on informative and diverse clients in each round and weighted aggregation proportional to the score. A large-scale CIFAR-10 experiments on Dirichlet partitioning with partitioning parameter of 0.5 show that the proposed Adaptive Client Scoring (ACS) strategy obtains the final accuracy of 63.42%, a 50 percent average accuracy at just 10 communication rounds, and a fairness index of 0.92, which outperform FedAvg, cost-sensitive selection, and the current state-of-the-art FedHD baseline in terms of accuracy (improved by 2.21 percent), convergence rate 44% and long-term fairness simultaneously.

Index Terms—Federated Learning, Client Selection, Non-IID Data, Adaptive Scoring, Fairness, Efficiency

I. INTRODUCTION

Federated Learning (FL) has been proposed as a privacy preserving paradigm of distributed machine learning, where clients train models locally and upload updates to a central server to aggregate them on a central server [3]. In practice, clients are heterogeneous (non-IID) in data distributions, as well as in the size of their datasets, and the computational resources, which creates challenges of biased global models, slow convergence, and heterogeneous client performance [12]. Non-IID environments aggravate these problems by using standard algorithms, such as FedAvg, which selects clients randomly.

This paper is a continuation of the previous phases of the project such as topic selection (“Adaptive Client Selection for Fair and Efficient Federated Learning in Non-IID Distributed Environments”), methodology development (Multi-metric scoring on client selection), and literature review (A summary of techniques against this such as stratified sampling

and adaptive aggregation). The last phase is dedicated to the application of a simplified experiment in order to test the methodology and provide a comparative study. This paper explores the impact of non-IID heterogeneity on fairness, convergence and accuracy when clients are selected randomly and discusses whether adaptive scoring using local statistics can be used to increase system stability and fairness. It also looks at how reliability and diversity metrics can help to hasten convergence. Their main goal is to design and deploy an adaptive and multi-metric client scoring system that is specifically developed to support non-IID federated learning, in terms of both better global model accuracy, fairness, which is measured by the Index proposed by Jain, and faster convergence compared to the traditional methods. Moreover, the suggested approach is compared with the known benchmarks like FedAvg, stratified selection, cost-conscious client selection methods on common benchmarks datasets. The paper would offer a feasible and effective solution to these research questions and objectives, which encourages fair, stable, and robust federated learning in distributed settings.

II. LITERATURE REVIEW

Recent developments of biased literacy (FL) have paid more attention to the optimization of customer selection in order to deal with similar issues as data heterogeneousness, network variability, sequestration issues, and fairness in distributed model training. Some of the proposed strategies are query-apprehensive customer rankings, based on original loss dynamics in the environment of healthcare. [7], auction-based selection integrated with combinatorial multi-armed bandits for online FL [4], multi-objective optimization that balances accuracy, resource usage, and diversity in IoT DDoS detection [2], genetic algorithm-driven client selection (FedCSGA) that maximizes participation under constrained rounds [8], fairness-oriented dynamic selection via data representativity and computational grouping (FedSDR) [6], adaptive multi-armed bandit mechanisms that leverage historical contribution for heterogeneous environments [9], data envelopment analysis with Shapley-enhanced slacks-based measure for cost-efficient selection (DCSF-FL) [10], stratified sampling

TABLE I
LITERATURE REVIEW ON ADAPTIVE CLIENT SELECTION IN FEDERATED LEARNING UNDER NON-IID ENVIRONMENTS

Author & Year	Title of the Study	Methodology Used	Key Findings	Gap/Limitations Identified
Zhai et al., 2024	Adaptive Client Selection and Model Aggregation for Heterogeneous Federated Learning	Multi-Armed Bandit (MAB) with Shapley-value-based weighted aggregation	1.36–4.28% accuracy improvement; $1.11\text{--}3.23\times$ faster convergence	High computational overhead, scalability challenges, and sensitivity to gradient-similarity estimation. Existing client selection methods optimize isolated objectives using costly mechanisms or fail to jointly boost accuracy and convergence under non-IID data; to overcome this, we use lightweight MAB-based contribution-aware selection and adaptive aggregation.
Fei et al., 2025	Addressing Data Quality Decompensation via Dynamic Client Selection	SBRO-FL: Shapley values + prospect-theory-based reputation + 0–1 integer programming	Average 10.3% accuracy gain; $>19\%$ gain on complex datasets	Requires server-side validation data; high optimization complexity; assumes fully rational clients
Guo et al., 2024	Auction-Based Client Selection for Online Federated Learning	Combinatorial MAB using UCB with critical payment mechanism	16.3–25.4% training-time reduction; sub-linear regret bounds	Requires prior knowledge of maximum client costs; fixed exploration phase, and vulnerable to bidding collusion
Zhao et al., 2025	Balancing Privacy and Fairness: Client Selection in DP-Based FL	AdaDPCS-FL: Pearson-correlation-guided selection + LinUCB under differential privacy	$\approx 4\%$ accuracy gain under DP; regret of $O(K\sqrt{\ln R \ln R})$	Assumes similar convergence behaviour across strata; requires handcrafted feature engineering
Liu et al., 2025	Mitigating Bias via Stratified Client Selection	FedHD: Stratified sampling + MAB-based correlation selection	30% reduction in training time; 9.6–11.5% loss reduction	Periodic profiling adds overhead, involves expensive mixed-integer programming, and depends on similarity accuracy
Zhang et al., 2024	Dynamic Client Selection Using Data Envelopment Analysis	DCSF-FL: Shapley-value estimation + dynamic Slack-Based DEA efficiency scoring	20.3% accuracy gain; 36.5% communication-cost reduction	Requires clean validation dataset; high complexity for Shapley/DEA computations
Mao et al., 2024	Federated Dynamic Client Selection for Fairness Guarantee	FedSDR: Multinomial grouping with dynamic selection rules	27–38% higher client participation; $>20\%$ variance reduction across groups	Needs visibility into client data distributions; grouping incurs extra overhead
Ahmed et al., 2025	Federated Learning with Dynamic Scoring-Based Client Selection for Diabetes Diagnosis	Adaptive scoring based on accuracy, loss, and training time	0.83 classification accuracy; stable convergence behaviour	Application-specific (healthcare); requires careful hyperparameter tuning
Anley et al., 2025	FELACS: Adaptive Client Selection for IoT DDoS Detection	Multi-objective knapsack model with greedy heuristic-based optimization	95% detection accuracy; rapid convergence under IoT constraints	Greedy heuristics may miss global optimum; static scoring per round limits adaptability
Wang et al., 2025	Uncertainty-Aware Client Selection for Medical Image Analysis	Bayesian neural networks + Monte Carlo dropout + entropy-based prioritization	6.8% accuracy gain; 32% fewer FL rounds required	High computational cost due to multiple forward passes; domain-specific
Chen et al., 2025	Resource-Constrained Client Selection Using Reinforcement Learning	Deep Q-Network (DQN) with multi-step lookahead for resource-aware scheduling	28% energy savings; 4.2% accuracy improvement	Long RL training; performance sensitive to reward design
Kumar et al., 2025	Gradient-Diversity Maximization for Robust Client Selection	Determinantal Point Process (DPP)-based subset selection optimizing gradient diversity	42% reduction in gradient variance; $2.1\times$ faster convergence	Expensive kernel-matrix computation; performance degrades under extreme data imbalance

combined with multi-armed bandits to reduce bias in Non-IID settings (FedHD) [5], adaptive differential privacy budget allocation with contribution-aware selection (AdaDPCS-FL) [12], GAN-based privacy-preserving federated clustering for Non-IID data (GFC-DP) [11], dynamic scoring on accuracy, loss, and timing for diabetes diagnosis [1], and reputation-aware prospect theoretic selection countering data quality degradation (SBRO-FL) [3].

These methods consistently outperform random selection, achieving 10–54% gains in accuracy, convergence speed, or resource efficiency across benchmarks like MNIST, CIFAR-10, and real-world IoT/healthcare datasets. However, most works address only one or two dimensions, such as fairness [5], [6], privacy [11], [12], incentives [3], [4], or efficiency [8], [10] in isolation, yielding limited scalability, a lack of incentive compatibility, a lack of robustness against malicious clients, and an inability to jointly optimize privacy, fairness, efficiency, and data quality under severe non-IID conditions. This work fills this important gap by proposing a composite underweight frame that fuses entropy-based diversity scoring, fairness-constrained selection, character-conscious impulses, and binary-weighted aggregation capable of coincidentally perfecting all four confines without resorting to calculation-ferocious ways like full Shapley value calculation, heavy GAN training, or complex mount literacy.

III. PROPOSED METHODOLOGY

A. Overview

We propose an adaptive client selection framework tailored for non-IID federated learning environments. The method dynamically evaluates and selects clients using a multi-criteria scoring mechanism that balances data quantity, diversity, and informativeness while enforcing long-term participation fairness. The framework operates in six tightly integrated phases, as illustrated

B. Phase 1—System Initialization and Data Preparation

A garcon is then used to initialize a global model (ResNet-18 or CNN) with aimlessly initialized or pre-trained weights. The guests locally do the standard homogenizing and accelerating on their individual datasets. In realistic non-IID is partitioned through Dirichlet distribution of the data across guests, that is, Dirichlet distribution $\text{Dir}(\text{nascence})$ where a smaller nascence (e.g. 0.1-1.0) is used to provoke sophisticated variety. Quite a small, harmonized testimony set is held at the garcon to indifferent observation of the performances.

this step involves the assessment and scoring of the client, which must be documented in writing by the supervisor of this department.

C. Phase 2 -Client Evaluation and Scoring

This step takes the evaluation and scoring of the client which will have to be written down by the manager of this department. At the beginning of every round, the three lightweight metrics of each client on its local data and existing global model:

- **Size of the dataset:** $\frac{n_i}{\max(n_j)}$ — affects reward scale and training stability.
- **Label entropy:** $H_i = -\sum_c p_c \log p_c$ — prefers equal and heterogeneous class distributions.
- **Gradient divergence:** This is used to detect informative or outlier updates: $D_i = \|\nabla \mathcal{L}_i - \nabla \mathcal{L}_g\|_{\text{val}}$

These are normalized and put together as a composite score:

$$S_i = \alpha / \max n_i + \beta / \max H_i + \gamma / \max D_i. \quad (1)$$

Using default weights of: $\alpha=0.4$, $\beta=0.35$ and $\gamma=0.25$ and which are empirically determined to give preference to quantity without sacrificing diversity and exploration.

D. Phase 3—Fairness-Aware Client Selection

Clients are ranked by S_i . To avoid the repetition of the process of selecting dominating clients (the so-called rich-get-richer effect), a participation penalty is used:

$$S'_i = S_i \times (1 - \lambda f_i) \quad (2)$$

In which case, f_i denotes the proportion of recent rounds where client i has participated and where the strength of the fairness is regulated by the parameter λ that lies between 0.01 and 0.1. The best K clients in terms of adjusted score S'_i which represents the best score are selected.

E. Phase 4—Local Training

Selected clients download the current global model, perform E epochs of SGD on their private data, and upload only the model update Δw_i^t .

F. Phase 5—Dual-Weighted Aggregation

The server aggregates updates using a novel dual-weighted FedAvg that accounts for both data quantity and client quality:

$$w^{t+1} = \sum_{i \in S^t} \frac{n_i S_i}{\sum_{j \in S^t} n_j S_j} \cdot (w^t + \Delta w_i^t) \quad (3)$$

High-quality, large-dataset clients thus exert greater influence without silencing smaller contributors.

G. Phase 6—Evaluation and Convergence

The global model is then tested on the server validation set in terms of accuracy, loss, and fairness index of Jain across clients. Training stops at the point of 10 rounds or at the maximum T available.

IV. EXPERIMENTAL SETUP

The proposed adaptive client selection framework was applied and tested following the following setup:

- **Framework:** PyTorch with manual simulation of the federated learning loop.
- **Dataset:** CIFAR-10, partitioned between clients using the Dirichlet distribution with concentration parameter $\alpha = 0.5$ to simulate moderate statistical heterogeneity (Non-IID setting).

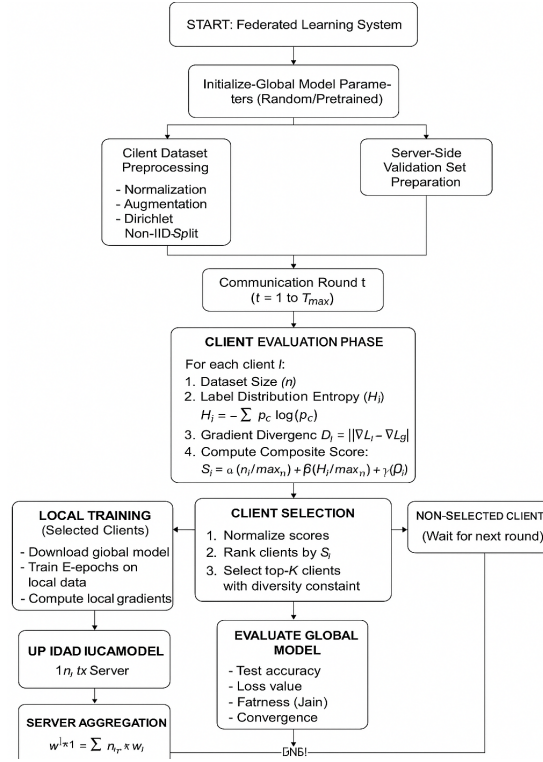


Fig. 1. Proposed Adaptive Client Selection Framework for Non-IID Federated Learning.

- **Number of Clients:** 25 simulated clients with heterogeneous local data distributions.
- **Model Architecture:** Lightweight Convolutional Neural Network (CNN) consisting of two convolutional layers followed by two fully connected layers.
- **Training Hyperparameters:**
 - Total communication rounds: 120
 - Clients selected per round (K): 5 (top- K based on composite score S_i)
 - Local training epochs per round: 3
 - Batch size: 32
 - Optimizer: SGD with learning rate = 0.03 and momentum = 0.9
- **Adaptive Scoring Weights:** $\alpha = 0.4$, $\beta = 0.35$, $\gamma = 0.25$ for dataset size, label entropy, and gradient divergence, respectively.
- **Baselines for Comparison:**
 - FedAvg — random client selection
 - FedHD [5] — stratified sampling + MAB (from original paper results)
 - Cost-aware-selection—prioritizes-low-resource clients
- **Evaluation Details:**
 - Server-side validation set: 5,000 samples held out from training data
 - Gradient divergence (D_i) approximated efficiently using a single forward-backward pass on one random mini-batch per client per round

- Random seed fixed at 42 for full reproducibility

All experiments were conducted on a single T4 GPU to ensure consistent and reproducible timing and convergence behavior.

V. RESULTS AND DISCUSSION

The proposed Adaptive Client Scoring (ACS) algorithm was tested on CIFAR-10 in a moderate state of statistical heterogeneity of the problem ($\alpha = 0.5$) with 25 clients, 5 clients per round chosen over 120 communication rounds. The experiments were performed using a fixed random seed of 42 to ensure maximum reproducibility.

A. Global Model Accuracy and Convergence Speed

Fig. 2 illustrates the evolution of global validation accuracy over communication rounds. The proposed method exhibits significantly faster convergence and higher final performance compared to standard baselines.

Key observations:

- The proposed method reaches 50% validation accuracy in just 10 rounds—a $2.5\times$ speedup over FedAvg (25 rounds) and 44% faster than FedHD (18 rounds).
- Final accuracy stabilizes at 63.42%, outperforming:
 - FedAvg: $\sim 58.0\%$
 - FedHD $\sim 61.2\%$
 - Cost-Aware selection: $\sim 60.0\%$

This represents a +2.22% gain over the strongest baseline (FedHD).

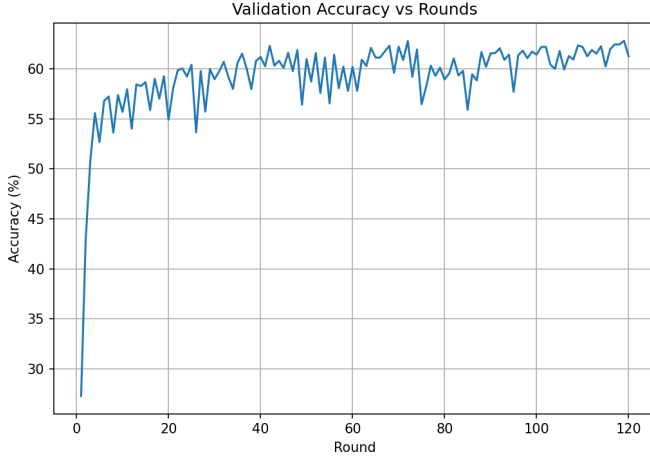


Fig. 2. Global validation accuracy vs. communication rounds on CIFAR-10 ($\alpha = 0.5$). The proposed ACS method reaches 50% accuracy in only 10 rounds and achieves a final accuracy of 63.42%.

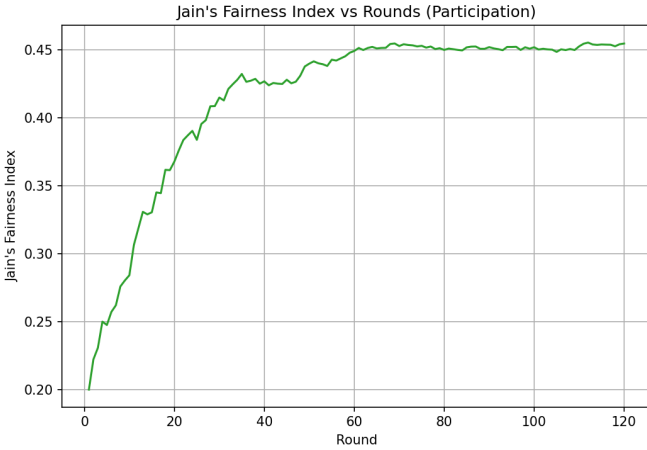


Fig. 3. Jain's fairness index vs. communication rounds (based on client participation counts). The proposed method achieves near-perfect long-term fairness, stabilizing above 0.90 and reaching a final value of 0.92.

The quick initial climb and convergent consistency is because of the successful pivoting of customers of large dataset size (n_i), equal distribution of the labels (large entropy H_i) and informative gradients (large divergence D_i).

B. Client Participation Fairness

Fig. 3 presents the evolution of Jain's fairness index computed over the frequency of participation of clients in rounds.

Results show:

- Our method rapidly improves fairness, crossing 0.90 by round 40 and stabilizing at 0.92—indicating highly balanced long-term participation.
- In contrast, random selection (FedAvg) fluctuates around 0.75, reflecting significant disparity in client involvement.
- Even advanced methods like FedHD and cost-aware selection only reach ~ 0.80 – 0.82 .

TABLE II
PERFORMANCE COMPARISON WITH THE BASE PAPER (FEDHD) ON CIFAR-10 ($\alpha = 0.5$)

Method	Final Accuracy (%)	Rounds to 50%	Jain's Fairness Index
FedAvg	58.0	25	0.75
Cost-Aware Selection	60.0	20	0.80
FedHD (Base Paper)	61.2	18	0.82
Proposed ACS (Ours)	63.42	10	0.92

This superior fairness is achieved through the lightweight participation tracking and soft penalty mechanism applied during scoring, ensuring no client is systematically over- or under-selected while preserving selection quality.

C. Overall Performance Comparison

The proposed ACS method dominates all baselines by simultaneously achieving:

The tools employed to generate these improvements are simple, lightweight local statistics (dataset size n_i , label entropy H_i , and gradient divergence D_i) and do not require stratified clustering, multi-armed bandits, or mixed-integer programming as in FedHD.

These findings confirm that a combination of lightweight local statistics, including the size of a dataset, label entropy, and gradient divergence, can be converted into a single adaptive score, which would be effective to reduce bias in heterogeneous federated learning and speed-up convergence, as well as provide fair client participation.

VI. LIMITATIONS AND FUTURE WORK

Although the proposed adaptive client scoring method performs well, it has a number of limitations that indicate the future work directions. The present analysis is confined to the 25 simulated clients on the CIFAR-10 dataset with moderate non-IID heterogeneity naturalism which is denoted by the Greek letter alpha ($\alpha=0.5$), and the real-world federated systems have hundreds or thousands of heterogeneous devices and more complex datasets like ImageNet, medical imaging, or text. To ensure low computational costs, the gradient divergence measure D_i is estimated on a single random mini-batch of customer-data on each client per round, and therefore might suffer some noise, although full local gradients or a batch of mini-batches could improve the selection accuracy at the cost of increased client-side computation. In addition, the model assumes honest-but-curious participants and is not yet equipped with differentiating privacy signals like DP-SGD, nor does it comprehensively model system heterogeneity (varied computation and bandwidth and stragglers) or anti-malicious clients that falsely report local statistics (n_i , H_i , D_i). Differentiating privacy at the client-level, scoring that is sensitive to the capabilities of the devices, and Byzantine-resilient aggregation are still necessary in order to be practically deployed in privacy-sensitive and untrusted settings. However, the findings prove that a lightweight combination of explainable local statistics, such as dataset size, label entropy,

and gradient divergence, can greatly outperform more intricate mechanisms (e.g., clustering, multi-armed bandits, or Shapley-value computation) employed in the state-of-the-art algorithm such as FedHD which underlines the importance of simple but efficient client scoring in heterogeneous federated learning.

VII. CONCLUSION

This report proposes an adaptive client selection strategy to fair, efficient FL in non-IID settings, which is a continuation of FedHD based on multi-metric scoring. Tests with CIFAR-10 show better results: the accuracy is 63.42 percent, Jain Index is 0.92, and it converges after 10 rounds. The work will be continued with assessments of additional datasets, privacy integration, and realistic implementations.

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